

Boss Challenge — Paper 47 (Class 7)

Heat | Electric Current & its Effects | Simple Equations | Symmetry

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. Heat always flows on its own from a hotter body to a colder one until both reach the:
(A) colder body's freezing point
(B) same temperature (thermal balance)
(C) hotter body's boiling point
(D) temperature of the Sun
2. When an electric current passes through a wire, the wire becomes hot. This is known as the:
(A) heating effect of electric current
(B) magnetic effect of electric current
(C) lighting effect of current
(D) chemical effect of electric current
3. A mathematical statement that two expressions are equal, using an equals sign, is called:
(A) a fraction
(B) an inequality
(C) an equation
(D) a ratio
4. A figure that can be folded along a line so that the two halves fall exactly on each other is said to have:
(A) no symmetry at all
(B) line (reflection) symmetry
(C) rotational symmetry only
(D) translation symmetry
5. The everyday unit used to state how hot or cold something is, written like 37°C , is the:
(A) metre
(B) kilogram
(C) degree Celsius
(D) second
6. An electric bulb lights up because the current heats its thin filament until it becomes:
(A) cold and shiny
(B) filled with coloured water
(C) magnetic and attracts iron
(D) white-hot and glows, giving out light
7. The value of the variable that makes both sides of an equation equal is called its:
(A) coefficient
(B) variable
(C) solution (root)
(D) constant

8. Counting both diagonals and both midlines, how many lines of symmetry does a square have?

- (A) 4
- (B) 8
- (C) 1
- (D) 2

9. The thermometer a doctor uses to measure body temperature, marked roughly from 35°C to 42°C, is the:

- (A) laboratory thermometer
- (B) maximum-minimum thermometer
- (C) clinical thermometer
- (D) kitchen oven thermometer

10. A safety device made of a wire with a low melting point that melts and breaks the circuit when the current grows too large is a:

- (A) fuse
- (B) battery
- (C) switch
- (D) bulb

11. Solving the equation $x + 9 = 16$ (find the value of x), you get:

- (A) $x = 16$
- (B) $x = 25$
- (C) $x = 7$
- (D) $x = 9$

12. The number of lines of symmetry in an equilateral triangle is:

- (A) 3
- (B) 2
- (C) 0
- (D) 1

13. On a clinical thermometer, a healthy person's normal body temperature reads close to:

- (A) 0°C
- (B) 50°C
- (C) 37°C
- (D) 100°C

14. When a compass is brought near a wire carrying a current, its needle moves (deflects). This shows that an electric current has a:

- (A) sound effect
- (B) magnetic effect
- (C) heating effect
- (D) freezing effect

15. Solving the equation $4x = 28$ for x gives:

- (A) $x = 32$
- (B) $x = 112$
- (C) $x = 24$
- (D) $x = 7$

16. The number of lines of symmetry in a rectangle (that is not a square) is:

- (A) 2
- (B) 1
- (C) 4
- (D) 0

17. The transfer of heat through a solid metal rod, passed on from particle to particle without the particles travelling, is called:

- (A) convection
- (B) evaporation
- (C) radiation
- (D) conduction

18. A coil of wire wound on a soft iron core that becomes a magnet only while current flows through it is called an:

- (A) simple fuse
- (B) electromagnet
- (C) dry cell
- (D) permanent magnet

19. Solving the equation $x/5 = 6$ for x gives:

- (A) $x = 5$
- (B) $x = 1.2$
- (C) $x = 11$
- (D) $x = 30$

20. Drawing every possible diameter, how many lines of symmetry does a full circle have?

- (A) 0
- (B) 4
- (C) infinitely many
- (D) 1

21. In liquids and gases, heat is carried mainly by the actual movement of the warmed fluid itself. This way of transferring heat is:

- (A) reflection
- (B) conduction
- (C) radiation
- (D) convection

22. In an electric bell, the device that becomes magnetic and pulls the iron strip to strike the gong is the:
- (A) a heating coil
 - (B) electromagnet
 - (C) a permanent bar magnet
 - (D) a glass fuse
23. Solving the equation $2x + 3 = 11$ for x gives:
- (A) $x = 7$
 - (B) $x = 4$
 - (C) $x = 8$
 - (D) $x = 5.5$
24. The number of lines of symmetry in the capital letter H is:
- (A) 1
 - (B) 2
 - (C) 0
 - (D) 4
25. The heat of the Sun reaches the Earth across empty space (no air in between). This transfer without any medium is:
- (A) radiation
 - (B) conduction
 - (C) boiling
 - (D) convection
26. A battery is best described as:
- (A) a single electric cell on its own
 - (B) a wire that melts on overload
 - (C) two or more electric cells joined together
 - (D) a type of switch
27. When the term '+ 6' is moved (transposed) from the left side to the right side of an equation, it becomes:
- (A) + 6 (sign unchanged)
 - (B) / 6
 - (C) $\times 6$
 - (D) - 6
28. The number of lines of symmetry in the capital letter A is:
- (A) 3
 - (B) 2
 - (C) 1
 - (D) 0

29. Metals such as copper and aluminium let heat pass through them easily. Such materials are called:

- (A) good conductors of heat
- (B) good insulators of heat
- (C) non-metals only
- (D) transparent to light

30. In a circuit diagram, the longer line of a cell's symbol stands for the:

- (A) negative (-) terminal
- (B) switch
- (C) positive (+) terminal
- (D) fuse

31. Solving the equation $5x - 2 = 13$ for x gives:

- (A) $x = 15$
- (B) $x = 2.2$
- (C) $x = 3$
- (D) $x = 11$

32. When turned once fully about its centre, a square matches itself how many times - its order of rotational symmetry?

- (A) 4
- (B) 2
- (C) 1
- (D) 8

33. Woollen clothes keep us warm in winter mainly because wool:

- (A) is a good conductor of heat
- (B) reflects sunlight onto the body
- (C) traps air, which is a poor conductor of heat
- (D) produces its own heat

34. If two bulbs are connected one after the other in a single loop (in series) and one bulb fuses, the other bulb will:

- (A) glow even brighter than before
- (B) go off too, because the circuit is broken
- (C) keep glowing as if nothing happened
- (D) turn into a magnet

35. "When a number is multiplied by 3, the result is 27." Writing $3x = 27$ and solving, the number is:

- (A) 81
- (B) 9
- (C) 24
- (D) 30

36. Turned once fully about its centre, an equilateral triangle matches itself how many times - its order of rotational symmetry?

- (A) 6
- (B) 3
- (C) 2
- (D) 1

37. During the day at the coast, the cool breeze that blows from the sea towards the land is called the:

- (A) monsoon wind
- (B) land breeze
- (C) sea breeze
- (D) trade wind

38. Three cells, each of 2 volts, are joined in series. Using the equation $V = 2 + 2 + 2$, the total voltage V they supply is:

- (A) 2 volts
- (B) 6 volts
- (C) 8 volts
- (D) 0 volts

39. "5 added to twice a number gives 17." Writing $2x + 5 = 17$ and solving, the number is:

- (A) 6
- (B) 8
- (C) 11
- (D) 12

40. The smallest angle through which a square must be turned to look exactly the same as before is:

- (A) 90°
- (B) 360°
- (C) 45°
- (D) 180°

41. Two identical metal cans, one painted dull black and one shiny white, are left in the Sun. The one that warms up faster is the:

- (A) dull black can
- (B) neither warms at all
- (C) shiny white can
- (D) both warm at exactly the same rate

42. A torch holds some 1.5-volt cells in series and the bulb works at 6 volts. Solving $1.5 \times n = 6$, the number of cells n needed is:

- (A) 3 cells
- (B) 2 cells
- (C) 4 cells
- (D) 6 cells

43. Solving the equation $4(x - 2) = 12$ for x gives:

- (A) $x = 14$
- (B) $x = 1$
- (C) $x = 3$
- (D) $x = 5$

44. The number of lines of symmetry in a scalene triangle (all sides different) is:

- (A) 0
- (B) 1
- (C) 2
- (D) 3

45. Cooking pans are given handles made of wood or plastic because these materials are:

- (A) good conductors of heat
- (B) able to make their own heat
- (C) heavier than the metal pan
- (D) poor conductors (insulators) of heat

46. To make an electromagnet stronger, the most effective single change you can make is:

- (A) increasing the number of turns in the coil
- (B) using fewer turns of wire
- (C) switching the current off
- (D) removing the iron core

47. Solving the equation $16 = x + 7$ for x gives:

- (A) $x = 7$
- (B) $x = 16$
- (C) $x = 9$
- (D) $x = 23$

48. The number of lines of symmetry in an isosceles triangle (exactly two equal sides) is:

- (A) 2
- (B) 3
- (C) 1
- (D) 0

49. The liquid inside a common thermometer rises up the thin tube when heated because the liquid:

- (A) changes colour with heat
- (B) shrinks on heating
- (C) turns into a gas
- (D) expands on heating

50. An electromagnet is wound with 50 turns of wire. To triple its turns you add x more, and $50 + x = 150$. Solving for x , you must add:
- (A) 200 turns
 - (B) 150 turns
 - (C) 100 turns
 - (D) 50 turns
51. Solving the equation $2x = -8$ for x gives:
- (A) $x = -4$
 - (B) $x = -16$
 - (C) $x = -6$
 - (D) $x = 4$
52. The number of lines of symmetry in a regular pentagon (5 equal sides) is:
- (A) 10
 - (B) 5
 - (C) 1
 - (D) 4
53. A clinical thermometer has a small kink (a narrow bend) just above the bulb. Its job is to:
- (A) make the liquid rise faster
 - (B) change Celsius into Fahrenheit
 - (C) stop the liquid from slipping back so you can read the value after removing it
 - (D) store extra liquid for high fevers
54. When the switch in a simple circuit is moved to the 'OFF' position, the circuit becomes:
- (A) hotter, so the bulb glows brighter
 - (B) magnetic, so it attracts iron
 - (C) closed, so more current flows
 - (D) open, so no current flows
55. A pan of water at 18°C is heated by a rise of r degrees to reach 70°C . Writing $18 + r = 70$ and solving, the rise r is:
- (A) 88°C
 - (B) 18°C
 - (C) 70°C
 - (D) 52°C
56. Which of these capital letters has BOTH line symmetry and rotational symmetry (of order more than 1)?
- (A) F
 - (B) O
 - (C) P
 - (D) R

57. A laboratory thermometer is usually marked over a wide range such as:

- (A) 200°C to 500°C
- (B) 0°C to 5°C only
- (C) -10°C to 110°C
- (D) 35°C to 42°C

58. Compared with a filament bulb, a CFL or LED is preferred for lighting because it:

- (A) makes much more heat than a filament bulb
- (B) needs no electricity at all
- (C) works only in sunlight
- (D) gives the same light while wasting far less energy as heat

59. A torch's bulb runs at 9 volts using cells of 1.5 volts in series. Writing $1.5n = 9$ and solving, the number of cells n is:

- (A) 9 cells
- (B) 3 cells
- (C) 4 cells
- (D) 6 cells

60. Saying that a shape has reflection symmetry is just another way of saying it has a:

- (A) right angle
- (B) mirror line (line of symmetry)
- (C) pair of parallel sides
- (D) centre of rotation

61. You should NOT use a clinical thermometer to measure the temperature of boiling water because:

- (A) boiling water is colder than the body
- (B) the kink melts in warm water
- (C) clinical thermometers cannot touch water at all
- (D) its scale stops near 42°C, far below water's boiling point of 100°C

62. Modern homes often use an MCB (Miniature Circuit Breaker) in place of a fuse because an MCB:

- (A) never breaks the circuit at all
- (B) turns the house lights into magnets
- (C) supplies extra current to the house
- (D) can be switched back on after it trips, instead of being replaced

63. Solving the equation $3x + 2x = 25$ for x gives:

- (A) $x = 25$
- (B) $x = 5$
- (C) $x = 12.5$
- (D) $x = 10$

64. Held up to a plane mirror, which of these capital letters looks exactly the SAME as its mirror image?

- (A) L
- (B) M
- (C) G
- (D) F

65. On a sunny day you stay cooler in white clothes than in dark ones because light-coloured cloth:

- (A) blocks all light from reaching you
- (B) absorbs more heat than dark cloth
- (C) makes its own cool air
- (D) reflects most of the heat falling on it

66. The hands-experiment in which a compass needle near a current-carrying wire is deflected was the discovery that linked:

- (A) heat and light
- (B) electricity and sound
- (C) water and electricity
- (D) electricity and magnetism

67. Solving the equation $x/3 = 4$ for x gives:

- (A) $x = 7$
- (B) $x = 4$
- (C) $x = 12$
- (D) $x = 1.3$

68. The number of lines of symmetry in a rhombus (a slanted diamond with 4 equal sides) is:

- (A) 2
- (B) 0
- (C) 1
- (D) 4

69. The temperature of some water rises from 25°C to a final value of $t^{\circ}\text{C}$, a total rise of 30°C .

Writing the equation $25 + 30 = t$, the final temperature t is:

- (A) 30°C
- (B) 5°C
- (C) 55°C
- (D) 750°C

70. The fuse wire in a household fuse is deliberately made of a metal that has a:

- (A) strong magnetism
- (B) very high melting point
- (C) low melting point
- (D) bright colour when cold

71. Solving the equation $2(x + 3) = 16$ for x gives:

- (A) $x = 5$
- (B) $x = 2$
- (C) $x = 11$
- (D) $x = 8$

72. The order of rotational symmetry of a rectangle (that is not a square) is:

- (A) 3
- (B) 1
- (C) 4
- (D) 2

73. A metal block is heated so its temperature goes from $x^{\circ}\text{C}$ up by 40°C to reach 95°C . Solving $x + 40 = 95$, the starting temperature x was:

- (A) 135°C
- (B) 55°C
- (C) 95°C
- (D) 40°C

74. An electric heater and an electric iron both work mainly using the current's:

- (A) heating effect
- (B) magnetic effect
- (C) cooling effect
- (D) chemical effect

75. Is $x = 2$ a solution of the equation $3x + 1 = 7$? Checking by substitution:

- (A) no, because $3 \times 2 + 1 = 9$
- (B) yes, because $3 \times 2 + 1 = 7$
- (C) cannot be checked without a graph
- (D) no, because x must be 7

76. What is the least angle of turn that brings an equilateral triangle back to looking the same?

- (A) 360°
- (B) 120°
- (C) 90°
- (D) 60°

77. A bimetallic strip (two different metals joined together) bends when heated because the two metals:

- (A) lose their colour
- (B) turn into liquid metal
- (C) expand by different amounts
- (D) both refuse to expand at all

78. Why is the heating element (coil) of a room heater made of a special wire rather than ordinary copper wire?
- (A) it is a magnet that attracts the room's dust
 - (B) it makes the current flow backwards
 - (C) it carries current without ever heating up
 - (D) it resists the current more and so gets much hotter, without melting
79. Solving the equation $6x - 4 = 2x + 12$ for x gives:
- (A) $x = 4$
 - (B) $x = 8$
 - (C) $x = 2$
 - (D) $x = 16$
80. A parallelogram that is not a rectangle or rhombus has how many lines of symmetry?
- (A) 0
 - (B) 2
 - (C) 1
 - (D) 4
81. While water is heated in a pan, the hotter water at the bottom rises and cooler water sinks, forming:
- (A) electric currents
 - (B) convection currents
 - (C) conduction currents
 - (D) radiation beams
82. In a circuit drawn on paper, a small circle with a cross or a loop inside it is the standard symbol for a:
- (A) bulb
 - (B) cell
 - (C) switch
 - (D) battery
83. "Twice a number, decreased by 5, equals 9." Writing $2x - 5 = 9$ and solving, the number is:
- (A) 7
 - (B) 2
 - (C) 4
 - (D) 14
84. The number of lines of symmetry in a regular hexagon (6 equal sides) is:
- (A) 4
 - (B) 6
 - (C) 12
 - (D) 3

85. Tungsten metal is chosen for the filament of an electric bulb partly because, where heat is concerned, it has a:

- (A) very high melting point
- (B) habit of staying cold
- (C) strong smell when heated
- (D) very low melting point

86. An electromagnet loses its magnetism the moment the current is switched off. This makes it especially useful for:

- (A) storing electricity like a cell
- (B) lighting a room without a bulb
- (C) making a magnet that can never be switched off
- (D) picking up and then releasing iron objects on demand

87. Solving the equation $x + x + x = 21$ for x gives:

- (A) $x = 21$
- (B) $x = 7$
- (C) $x = 3$
- (D) $x = 63$

88. The capital letter S has no line of symmetry, yet it looks the same when turned upside down (180°). This means S has:

- (A) neither line nor rotational symmetry
- (B) line symmetry but no rotational symmetry
- (C) both line and rotational symmetry
- (D) rotational symmetry but no line symmetry

89. Heat from a room heater spreads to warm the whole room mainly by:

- (A) convection of the air
- (B) the air turning into water
- (C) light reflecting off the walls
- (D) conduction through the air

90. The energy change taking place in an electric bulb that is switched on is:

- (A) electrical energy into light and heat
- (B) sound energy into light
- (C) light energy into electricity
- (D) heat energy into magnetism

91. Solving the equation $10 - x = 4$ for x gives:

- (A) $x = 6$
- (B) $x = -6$
- (C) $x = 14$
- (D) $x = 4$

92. The number of lines of symmetry in the digit 8 (written in its symmetric printed form) is:

- (A) 0
- (B) 4
- (C) 1
- (D) 2

93. When you hold a metal spoon and a wooden spoon, both at room temperature, the metal one feels colder. This is because the metal:

- (A) reflects heat back into your hand
- (B) is actually at a lower temperature
- (C) makes its own cold
- (D) conducts heat away from your hand faster

94. A bird can perch safely on a single high-voltage wire because:

- (A) birds cannot feel electricity
- (B) the wire has no electricity in it
- (C) current needs a complete path, and the bird does not provide one to a different point
- (D) its feathers are made of metal

95. A pendulum makes some swings and 12 more make 40 in all. Writing $s + 12 = 40$ and solving, the first number of swings s was:

- (A) 12
- (B) 52
- (C) 40
- (D) 28

96. A heating coil is laid out so that its left half is a mirror image of its right half about a central vertical line. This central line is the coil's:

- (A) centre of rotation
- (B) line of symmetry
- (C) diagonal of a square
- (D) angle of incidence

97. At night by the sea, the breeze blows from the land towards the sea. This land breeze forms because, after sunset, the:

- (A) sea cools faster than the land
- (B) land cools faster than the sea
- (C) land makes its own wind
- (D) Sun heats the land at night

98. Which of these is the correct order of parts that lets a simple torch light up?

- (A) bulb alone with no cells or wires
- (B) cells joined to a switch that is left open
- (C) cells $\rightarrow$ switch (on) $\rightarrow$ bulb, joined in a complete loop by wires
- (D) two bulbs with no cells at all

99. Solving the equation $7x = 0$ for x gives:

- (A) $x = 7$
- (B) no solution exists
- (C) $x = 1$
- (D) $x = 0$

100. The number of lines of symmetry in a semicircle (half a disc) is:

- (A) infinitely many
- (B) 1
- (C) 2
- (D) 0